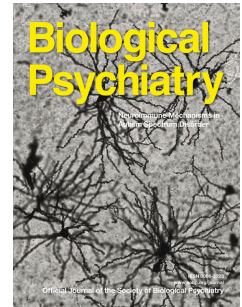


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Polygenic Risk for Major Depression: Diagnostic Specificity and Developmental Trajectories in an Admixed Youth Cohort

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Polygenic Risk for Major Depression: Diagnostic Specificity and Developmental Trajectories in an Admixed Youth Cohort

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ABSTRACT

Background: Major Depressive Disorder (MDD) is heritable and polygenic, yet the relative diagnostic specificity, developmental impact, and cross-ancestry generalizability of MDD polygenic risk scores (MDD-PRS) remain unclear.

Methods: In two sites of the Brazilian High-Risk Cohort (N=2,163; ages 6–23; 45.6% female), we used a discovery–replication design to test associations between MDD-PRS and (i) lifetime DSM-IV diagnoses; (ii) longitudinal depressive-symptom trajectories from latent growth-curve models; and (iii) late-adolescent/young-adult depressive symptoms and nonsuicidal self-injury (NSSI). Analyses used weighted ordinary least squares and quantile regression, adjusting for age, sex, socioeconomic status, genetic principal components, and psychiatric comorbidity.

Results: MDD-PRS showed relative specificity for MDD in both sites, explaining approximately 2.6–3.6% of liability-scale variance; no other diagnostic category survived false-discovery-rate correction. Longitudinally, higher MDD-PRS predicted higher initial level ($\beta=0.10$ – 0.18 ; $p<0.001$ in both sites), faster rate of increase ($\beta=0.07$ – 0.10 ; $p=0.004$ and $p<0.001$), and higher time-averaged level ($\beta=0.13$ – 0.20 ; $p<0.001$ in both sites) of depressive symptoms, with effects strengthening at higher quantiles. Cross-sectionally, MDD-PRS were associated with more depressive symptoms ($\beta=0.15$ – 0.17 ; $p<0.001$ in both sites). For NSSI, a main-effect association was observed ($\beta=0.09$; $p<0.001$), but it remained significant only in a post-hoc MDD-PRS $\times$ lifetime-MDD interaction among individuals with MDD ($\beta=0.24$; $p=0.014$).

Conclusions: In this deeply phenotyped, admixed cohort, MDD-PRS showed relative diagnostic specificity and marked a more severe developmental course of depression, with NSSI

associations contingent on MDD diagnosis, supporting the relative specificity, developmental utility, and cross-ancestry relevance of MDD-PRS.

INTRODUCTION

Major Depressive Disorder (MDD) is a leading cause of disability worldwide, with a significant proportion of its multifactorial etiology accounted for by genetic factors (1,2). For example, twin and family studies estimate the heritability of MDD to be approximately 37% (3). Genome-wide association studies (GWAS) show that this heritability reflects in part the cumulative effects of thousands of common genetic variants, each with a small effect (2,4). This polygenic architecture can be captured by polygenic risk scores (PRS), which aggregate risk variants into a single measure of an individual's genetic liability (5,6).

Despite this progress, key gaps remain. First, the relative diagnostic specificity of MDD-PRS is not fully established, as these scores may be associated with general psychopathology rather than MDD-specific risk (6–8). Second, relative specificity and cross-ancestry portability are rarely evaluated in the same design. Third, the developmental role of PRS is unclear, whether higher scores mark elevated baseline symptoms, faster symptom accumulation, or both. Recent longitudinal work indicates that genetic associations differ by symptom trajectory (9) and that PRS can distinguish youths likely to recover from those at risk for chronic depression (10). In diverse U.S. adolescents, MDD-PRS predict higher baseline symptoms and higher-risk trajectory classes across racial/ethnic groups, but rate-of-change effects were observed primarily in White youth (11). Whether MDD-specific PRS shape the developmental course—especially symptom change over time—beyond transdiagnostic liability and in admixed non-U.S. cohorts, remains underexplored.

Focusing only on the categorical diagnosis overlooks the heterogeneity of depression: different symptom measures yield different genetic associations (7,8) and can obscure how genetic risk maps onto dimensional severity and related clinical phenomena (12). This is especially critical in

youth, where these conditions first emerge and co-develop (13). Establishing whether MDD-PRS capture variance relatively specific to MDD therefore remains a necessary step.

Critically, most psychiatric genomics studies have been conducted in European-ancestry cohorts, limiting generalizability. Evaluating PRS performance in diverse, admixed populations is essential for equity and portability (14,15). In Latin American youth, European-derived depression PRS have predicted past-year MDD and major depressive episodes in Mexican adolescents (16), but such evidence is scarce, cross-sectional, and based on European-only weights. Latin American populations remain persistently underrepresented (17), and the performance of multi-ancestry MDD-PRS in Brazilian cohorts is largely unknown—despite nationwide whole-genome data now mapping fine-scale ancestry in Brazil (18). Moreover, the latest multi-ancestry MDD GWAS explained substantially less variance in non-European ancestries, and multi-ancestry weights did not clearly outperform European-only scores (2). These gaps motivate testing whether multi-ancestry MDD-PRS show relative diagnostic specificity and developmentally relevant associations in a highly admixed Brazilian cohort.

The present study advances the literature by (i) testing the relative diagnostic specificity of MDD-PRS for MDD while adjusting for comorbid psychiatric conditions; (ii) modeling developmental trajectories (baseline level, rate of change, cumulative burden) of depressive symptoms; and (iii) mapping heterogeneity across the symptom distribution with quantile regression. In doing so, we jointly assess relative diagnostic specificity and preliminary cross-ancestry utility across two sites of a highly admixed youth cohort. The primary outcome was lifetime MDD diagnosis. Secondary outcomes were (i) longitudinal trajectory parameters—intercept, slope, and time-standardized area under the curve (AUC/time)—derived from depressive/internalizing scales spanning childhood to young adulthood; and (ii) late

adolescence/young adulthood: depressive symptoms and nonsuicidal self-injury (NSSI), included as a clinically relevant correlate of depressive genetic liability rather than a genetically independent phenotype. We utilized data from two sites of the Brazilian High-Risk Cohort (BHRC) (19,20), a 15-year community school-based longitudinal study in a highly admixed population (21–23).

METHODS AND MATERIALS

Study design and participants

This study included participants from the BHRC, a community school-based longitudinal study with an enriched-risk design (i.e., oversampling for child and family psychopathology) conducted at two major Brazilian metropolitan sites: Porto Alegre and São Paulo (19,20).

Following a school-based screening phase, the baseline sample (Wave 0; 2010–2011) included 2,511 children and adolescents aged 6–14 years ($M = 10.20$, $SD = 1.90$). The first follow-up (Wave 1; 2014–2015) reassessed 2,010 participants (retention = 80.02%), aged 9–18 years ($M = 13.50$, $SD = 1.92$), and the second follow-up (Wave 2; 2018–2019) evaluated 1,905 participants (retention = 76.1%), aged 13–23 years ($M = 18.33$, $SD = 2.01$). Of these, our genotyped analytic sample (post-quality control) comprised 2,163 participants at baseline (86.1% of Wave 0), 1,776 at Wave 1 (88.4% of Wave 1 participants), and 1,690 at Wave 2 (88.7% of Wave 2 participants; Supplementary Figure S1).

We utilized data from all three waves for individuals with available MDD-PRS (see Results for sample characteristics). The analysis followed a five-stage discovery–replication workflow, with Porto Alegre arbitrarily designated as the discovery site before any data analysis and São Paulo as the replication site: (1) calculation of composite sampling×attrition weights to correct for the

oversampling design and attrition; (2) genotyping, quality control, and imputation, principal components and relatedness estimation, and construction of the MDD-PRS; (3) derivation of outcomes: categorical psychiatrist-confirmed lifetime Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV) (24) diagnoses, harmonized depressive-symptom trajectories across Waves 0–2, and latent depressive-symptom and NSSI scores at Wave 2; (4) diagnostic specificity of MDD-PRS and their association with cross-sectional and longitudinal outcomes; (5) pooling of regression estimates across multiple imputed datasets and bootstrap iterations within site (across sites for NSSI), and comparison of discovery and replication results (Supplementary Figures S2–S3). Reporting adhered to Strengthening the Reporting of Genetic Association Studies (STREGA) (25) and PRS Reporting Standards (PRS-RS) (26).

This study was approved by the Brazilian National Ethics in Research Commission (CAAE 13852413010015327 and 74563817.7.1001.5327). The participants' parents provided written informed consent, and all children provided assent (19,20).

Genetic Data Processing and Analysis

Genotyping, quality control and imputation. DNA was genotyped on the Illumina Global Screening Array (Illumina, San Diego, CA, USA) (~700,000 single nucleotide variants [SNVs]). Standard quality control (QC) excluded variants (illustratively, minor allele frequency [MAF]<1%, call rate<98%, Hardy–Weinberg equilibrium [HWE] $p < 1 \times 10^{-10}$) and samples with call rate<98% or sex discordance; imputation used TOPMed r3 (27), chosen for its ancestral diversity and representation of admixed populations, with retention of variants at imputation accuracy $R^2 > 0.8$. Final analytic N after QC/relatedness steps is given above. A complete description of QC thresholds, pre-imputation checks, phasing/imputation software and counts is provided in Supplement.

Principal component analysis, relatedness and ancestry estimation. Population structure and relatedness were handled with PC-AiR and PC-Relate (28,29); one individual per pair with kinship > 0.10 was removed, and the top 10 genetic principal components (PCs) were included as covariates in all models. Global ancestry was summarized with ADMIXTURE (K=4) using 1000 Genomes Projects and Human Genome Diversity Project (HGDP) references. Full implementation details and site-stratified ancestry visualizations are provided in the Supplementary Methods and Supplementary Figures S4–S5.

MDD Polygenic Risk Scores (MDD-PRS). MDD-PRS were derived from the Psychiatric Genomics Consortium (PGC) multi-ancestry GWAS (2) using PRS-CS-auto (30), a Bayesian continuous-shrinkage method that automatically estimates the global shrinkage parameter, with all other hyperparameters at recommended defaults. Scores were computed in PLINK (31) and z-standardized within each site. Z-standardization rescales PRS to mean zero and unit variance within each site, facilitating cross-site comparison of effect sizes; distributions closely approximate normality (Supplementary Figure S6). Software packages and the full weighting pipeline appear in the Supplementary Methods and Supplementary Table S1.

Major depressive disorder and other categorical diagnoses

Across waves, DSM-IV diagnoses were assessed with the Development and Well-Being Assessment (DAWBA) (32,33) and reviewed by trained psychiatrists to ascertain current MDD episodes and other diagnoses at each wave. Participants were then classified as lifetime cases for a given diagnosis if criteria were met at any of the three assessment waves. Raters at each wave were blinded to prior-wave assessments. For regression, diagnoses were grouped into five non-mutually exclusive categories to avoid data sparsity: MDD; anxiety disorders, attention-deficit hyperactivity disorder (ADHD), conduct disorder (CD) or oppositional defiant disorder (ODD),

and ‘other psychiatric disorders’. Instrument details and the full diagnosis list by site are in the Supplementary Methods and Supplementary Table S2.

Depressive Symptoms

Depressive symptoms were assessed with participant self-report using the Short Mood and Feelings Questionnaire (SMFQ) (34) at Wave 2, and via parent/main-caregiver reports at every wave using the Child Behavior Checklist (CBCL) for participants under 18 years of age and the Adult Behavior Checklist (ABCL) for adults (35). We derived an SMFQ latent factor via confirmatory factor analysis (CFA), harmonized CBCL/ABCL depressive items to compute a DSM-5–Oriented Depressive Problems latent factor, and estimated a longitudinal latent depressive-symptom factor using latent growth curve modeling (LGCM). Instrument administration and rater details are provided in the Supplement.

Nonsuicidal Self-Injury (NSSI)

NSSI at Wave 2 was measured with the Deliberate Self-Harm Inventory Nine-Item Version (DSHI-9) (36). A latent factor score was derived from CFA. Instrument administration and rater details are provided in the Supplement.

Covariates

All models were adjusted for age, sex assigned at birth, continuous socioeconomic status (37), and PCs 1–10. Diagnostic-specificity models were additionally adjusted for comorbidity (anxiety disorders, ADHD, CD/ODD, and other disorders; or MDD when a non-MDD diagnosis was the outcome).

Oversampling and attrition weights

Oversampling weights (the inverse of each participant's selection probability) restore population-representative estimates by correcting for the enriched-risk design of the BHRC. Inverse-probability-of-retention weights, derived from a logistic-regression model predicting continued participation from baseline covariates, aim to adjust for known variables potentially associated with non-random attrition. The final composite weight is the product of both components and was applied in all models (see Supplementary Methods and Supplementary Table S3).

Statistical Analysis

Descriptive analyses characterized the analytic sample and contrasted recruitment sites using the composite weights. All non-genetic analyses and visualizations were performed in R 4.5.0 (38).

Diagnostic specificity. For lifetime diagnoses, we fit site-specific weighted logistic regressions of each diagnostic category on the z-standardized MDD-PRS, adjusting for the covariates listed above, and included general comorbidity (presence of any diagnoses other than the one under analysis). Inference used a weighted bootstrap nested with multiple imputation (wBMI) (39), and Benjamini–Hochberg false discovery rate (FDR) correction was applied within site across diagnostic categories. We summarized variance explained with Nagelkerke's ΔR^2 ; for MDD we also report variance on the liability scale assuming 15% population prevalence (4,40) using the transformation of Lee et al. (41). Two sensitivity analyses re-estimated diagnostic models with (i) a cross-disorder PRS (42) as an additional covariate and (ii) EUR-ancestry-only MDD-PRS in place of the multi-ancestry PRS (see Supplementary Methods).

Dimensional outcomes. For longitudinal depressive symptoms, we tested scalar invariance of the harmonized CBCL/ABCL latent factor and used alignment optimization (43,44). We then fit weighted LGCMs with time loadings anchored to mean age gaps between waves; model

comparisons supported a parsimonious specification with quadratic-slope variance fixed to zero (Supplement). We report effects on the LGCM intercept, linear slope, and a time-averaged level (AUC/time). AUC/time is a linear combination of the intercept and slope, summarizing cumulative burden rather than independent information; all three are reported because they address complementary developmental questions despite their inherent correlation. For Wave-2 cross-sectional analyses, SMFQ and NSSI latent scores were regressed on the z-standardized MDD-PRS using weighted OLS and weighted quantile regression (QR) to characterize heterogeneity across the outcome distribution; NSSI models were pooled across sites and included a post-hoc MDD-PRS $\times$ lifetime-MDD interaction. Trends in QR effects across quantiles were tested for significance with a Wald-type trend test (45), and uncertainty was summarized with pointwise and simultaneous confidence bands (46). Full implementation details, QR grids and band construction, and replication criteria are provided in the Supplementary Method and Supplementary Figures S2–S3.

RESULTS

Sample, follow-up, and population structure

After QC and pruning (Supplement) of related individuals, the baseline analytic sample comprised 2,163 unrelated probands with valid MDD-PRS. Descriptive characteristics are shown in Table 1 and the participant flowchart in Supplementary Figure S1. Except for raw counts (N), all Table 1 values are adjusted using composite weights to account for oversampling and attrition. Sites were broadly similar in demographics and SES and expected ancestry differences due to historical migration patterns were observed (18): higher African ancestry and Mixed self-identification in São Paulo, higher European ancestry in Porto Alegre. These differences are visualized in Supplementary Figures S4–S5 (ADMIXTURE and site-stratified PC scatterplots).

Descriptive statistics for psychiatric outcomes

Mental-health outcomes by site are summarized in Table 2. Anxiety disorders were most prevalent, followed by MDD, ADHD, CD or ODD, and other disorders. Harmonized CBCL/ABCL depressive-problems scores increased across waves in both sites. Comorbidity was prevalent; among participants with lifetime MDD, 73.9% in Porto Alegre and 71.2% in São Paulo met criteria for at least one additional category (Supplementary Table S4).

Diagnostic-specificity of MDD-PRS

Figure 1 (panel A) shows the lifetime probability for each categorical diagnosis adjusted for psychiatric comorbidity in both sites. The results are expressed in average marginal effects (AME), which represent the increase or decrease in the probability (percentage-points) of the diagnosis per each unit of SD increase in MDD-PRS. MDD-PRS were associated with the highest AME for MDD, explaining ~2.6-3.6% of the variance on the genetic liability scale. No other diagnostic category survived FDR correction, which supports the relative diagnostic specificity of MDD-PRS for MDD at both sites. Adjusted probability of MDD across deciles of MDD-PRS is also shown in Figure 1 (panel B). After adjusting for comorbidities, the estimated probability of MDD in the lower tail of the MDD-PRS distribution (p15) was 2.8% (95% CI 2.0–3.7) for Porto Alegre and 2.5% (95% CI 1.8–3.3) for São Paulo; in the upper tail (p85) it was 6.9% (95% CI 5.4–9.2) for Porto Alegre and 6.4% (95% CI 4.8–9.1) for São Paulo. These findings were robust when adjusting for a cross-disorder PRS (Supplementary Figure S7) and were markedly attenuated when substituting EUR-ancestry-only MDD-PRS (Supplementary Figure S9).

Longitudinal depressive-symptom trajectories

A multi-wave CFA of CBCL/ABCL Depressive Problems showed good fit across waves (CFI = 0.964, TLI = 0.956, RMSEA = 0.037). Scalar invariance was supported across waves and we applied alignment optimization to the latent factors (Supplementary Table S5). The LGCM fit well ($\chi^2_{MV(1)} = 0.24$, $p = .623$; robust CFI/TLI = 1.000; RMSEA = 0.000; SRMR = 0.003) (Supplementary Table S6). Supplementary Figure S8 illustrates the between-person heterogeneity in CBCL/ABCL depressive-symptom trajectories captured by the model.

Figure 2 summarizes associations between MDD-PRS and LGCM parameters—initial level, rate of change, and average modeled level (AUC over time). OLS associations were positive and significant at both sites (initial level $\beta = 0.10$ – 0.18 , 95% CI 0.05 – 0.23 ; rate of change $\beta = 0.07$ – 0.10 , 95% CI 0.02 – 0.15 ; average level $\beta = 0.13$ – 0.20 , 95% CI 0.08 – 0.26 ; all $p \leq 0.004$). Quantile-regression trend tests were significant at both sites (Slope(τ) ranges: initial level 0.17 – 0.30 ; rate of change 0.27 at each site; average level 0.17 – 0.29 ; all $p < 0.001$). Under simultaneous confidence bands, the initial-level effect was significant in Porto Alegre ($\tau = 0.33$ – 0.89) but not in São Paulo; the rate-of-change effect was significant in the upper tail in both sites (Porto Alegre $\tau = 0.75$ – 0.92 ; São Paulo $\tau = 0.81$ – 0.97); and the average-level effect was broadly significant (Porto Alegre $\tau = 0.17$ – 0.93 ; São Paulo $\tau = 0.07$ – 0.20 and 0.85 – 0.99). Taken together, MDD-PRS relate to initial status and, more robustly, to faster symptom accumulation and higher average burden, with effects magnified toward higher quantiles and strongest replication for slope and average-level parameters.

Late adolescence/young adulthood dimensional outcomes

Single-wave (Wave 2) CFAs showed good fit: SMFQ (CFI = 0.995, TLI = 0.994, RMSEA = 0.033, SRMR = 0.032) and DSHI-9 (CFI = 0.986, TLI = 0.981, RMSEA = 0.022, SRMR = 0.070). Further details are provided in Supplementary Table S7. Figure 3 shows cross-sectional

associations between the MDD-PRS and depressive symptoms (SMFQ) at Wave 2. OLS effects were positive and significant in both sites (Porto Alegre: $\beta = 0.17$, 95% CI 0.08–0.25, $p < 0.001$; São Paulo: $\beta = 0.15$, 95% CI 0.08–0.24, $p < 0.001$). Quantile regression revealed significant effects across the middle-to-upper range of the symptom distribution in both sites (simultaneous confidence bands: Porto Alegre $\tau = 0.41$ –0.93; São Paulo $\tau = 0.50$ –0.98). A trend test indicating stronger effects at higher quantiles was significant in São Paulo (Slope(τ) = 0.23, $p = 0.007$) but not in Porto Alegre.

At Wave 2, 299 subjects endorsed NSSI, of whom 110 (36.8%) had MDD. As shown in Figure 4, in an analysis pooled across sites, higher MDD-PRS were associated with higher NSSI latent scores (main-effects model: OLS $\beta = 0.09$, 95% CI 0.04–0.15, $p < 0.001$), with quantile regression showing positive effects in the upper tail (simultaneous confidence bands significant for $\tau = 0.87$ –0.99) but no overall trend. In a post hoc interaction model testing MDD-PRS $\times$ lifetime MDD diagnosis, the association was significant for individuals with lifetime MDD ($\beta = 0.24$, 95% CI 0.04–0.43, $p = 0.014$), with pointwise—but not simultaneous—significance at upper quantiles ($\tau = 0.80$ –0.95) and no trend. Among those without lifetime MDD, the OLS model showed no association between MDD-PRS and NSSI; quantile regression showed no significant quantiles and no trend.

DISCUSSION

In this two-site Brazilian cohort, MDD-PRS showed relative diagnostic specificity for lifetime, psychiatrist-confirmed MDD and replicated across sites. Higher scores predicted higher initial levels, faster increases, and a greater average burden of depressive symptoms, with effects strongest toward the upper quantiles. Cross-sectionally, MDD-PRS related to more depressive symptoms, and to NSSI only among individuals with lifetime MDD.

Our study identified a specific genetic signal for MDD. Many large discovery GWAS rely on “minimal phenotyping,” which can dilute disorder-specific signals and emphasize a broader dysphoria construct (8,47). We addressed this by using deep phenotyping—DAWBA semi-structured assessments with psychiatrist-consensus diagnoses—and by modeling psychiatric comorbidity to isolate variance uniquely associated with MDD. Across both sites, the association between MDD-PRS and lifetime MDD persisted after these controls, supporting relative specificity beyond transdiagnostic liability.

These findings should be interpreted alongside the well-documented genetic overlap across psychiatric disorders, often conceptualized as a general psychopathology or p-factor (48). To address this, all diagnostic models were re-estimated including a cross-disorder PRS from the CDG3 analysis of 14 disorders (42). MDD-PRS remained significantly associated with MDD at both sites ($pFDR < 0.001$), no other category reached significance, and the dose–response gradient was preserved (Supplementary Figure S7), supporting relative—though not absolute—specificity above and beyond transdiagnostic genetic risk. Recent comorbidity-focused genetic studies similarly suggest that apparent genetic overlap between depression and conditions (e.g., ADHD, anxiety disorders) partly reflects the inclusion of comorbid cases rather than shared liability, while depression-related liability remains distinguishable (49,50).

Beyond relative diagnostic specificity, our longitudinal analyses provide a developmental view of how polygenic risk for depression unfolds. Higher MDD-PRS predicted higher initial depressive symptoms and faster accumulation through adolescence, with effects concentrated at higher quantiles—implying that genetic liability shapes the developmental course rather than conferring static risk. This complements and extends prior trajectory work. Grimes et al. (9) found that univariate MDD-PRS predicted persistent adolescent depression—although multitrait

(p-factor) scores were stronger and MDD-PRS showed no consistent association with trajectories in non-European ancestry groups tested in ABCD—and Jamil et al. (11) linked MDD-PRS to higher baseline symptoms and higher-risk trajectory classes in diverse U.S. adolescents, with limited rate-of-change effects observed primarily among White youth. Extending these, our disorder-specific multi-ancestry scores were associated with both upper-distribution status of depression and the rate of symptom acceleration in an admixed Brazilian cohort. Such a trajectory plausibly precedes the elevated risk of recurrent depression in adults with higher polygenic burden (51).

Because the LGCM intercept, slope, and AUC/time are inherently correlated, their individual PRS associations are not fully independent. However, the slope effect replicated across both sites and reached simultaneous-band significance at both, whereas the intercept did so only in Porto Alegre—suggesting that, despite their correlation, the slope captures partially distinct developmental information and provides the strongest evidence for a dynamic, trajectory-shaping influence of genetic risk.

The quantile-regression findings—stronger MDD-PRS effects at higher symptom quantiles—should be interpreted cautiously, as alternative explanations include floor effects in symptom measurement at lower quantiles, gene–environment correlation, and unmeasured confounders that concentrate at higher severity. Nonetheless, these findings extend OLS results by showing that genetic influence on depressive symptoms is heterogeneous across the severity distribution, a pattern relevant for risk-stratification approaches.

On the liability scale, MDD-PRS explained 2.62% (discovery) and 3.57% (replication) of variance using multi-ancestry GWAS weights, values comparable to ~4% in European-ancestry validation cohorts from the discovery GWAS (2) and above the ~1–2% typically reported in

independent samples (4,52). A sensitivity analysis substituting EUR-ancestry-only MDD-PRS showed substantially less variance explained (ΔR^2 liability $\approx 0.3\text{--}0.4\%$) and no significant associations at either site, with a markedly attenuated dose–response gradient (Supplementary Figure S9), demonstrating that multi-ancestry weights substantially outperform European-only weights in this admixed sample (53).

Beyond case–control prediction, MDD-PRS may help identify youths on steepening depressive-symptom trajectories, supporting earlier monitoring and preventive interventions during adolescence. This aligns with a recent multi-cohort survival analysis showing that MDD-PRS predicted earlier mood-disorder onset even after adjusting for family history and premorbid anxiety/ADHD (54). Our NSSI results indicate a gene-by-phenotype pattern: higher MDD-PRS related to greater NSSI burden only among individuals with lifetime MDD. Thus, pleiotropic effects of depression liability appear context-dependent—the clinical manifestation of MDD may activate or amplify pathways by which genetic risk contributes to self-injury—yielding a more severe, complex presentation once the primary disorder is established. This is consistent with prior work in the same cohort identifying two distinct psychopathology-based NSSI profiles, one of which is characterized by elevated, persistent comorbid psychiatric symptoms (55).

This study has several strengths and limitations. Strengths include testing associations in a deeply phenotyped, admixed Brazilian cohort recruited from two metropolitan sites (Porto Alegre and São Paulo), which addresses the European-ancestry bias in psychiatric genomics and provides preliminary evidence of cross-ancestry utility while adding to emerging Latin-American resources (e.g., state-level ancestry maps and regional reviews on portability) (17,18,56).

Methodologically, interview-based DAWBA assessments with psychiatrist-consensus diagnoses, together with longitudinal dimensional outcomes from harmonized CBCL/ABCL growth models

and independent Wave-2 instruments (SMFQ, DSHI-9), directly counter concerns about “minimal phenotyping.” A discovery–replication design across both sites, comorbidity adjustment, and composite sampling-plus-attrition weights further increase robustness in an admixed setting.

The effect sizes observed ($\beta \approx 0.07\text{--}0.24$) are modest and insufficient for individual-level clinical prediction, consistent with MDD's polygenic architecture—though our liability-scale estimates (2.6–3.6%) represent 31–43% of variance attributable to common SNPs, near the current methodological ceiling. The ~2.5-fold increase in MDD probability between the 15th and 85th PRS percentiles suggests meaningful population-level risk differentiation. Clinically meaningful individual-level stratification will likely require integrating PRS with environmental and clinical data, though the trajectory findings—identifying youth on accelerating symptom courses—may offer more actionable prognostic information than static associations.

Limitations include a developmental window restricted to childhood through young adulthood—so patterns, including symptom acceleration linked to polygenic risk, may not extend to first onset in mid- or late-life; a cross-sectional assessment of NSSI at the final wave, which precludes trajectory modeling; and high-risk oversampling, which may limit prevalence generalizability despite weighting. Additionally, although the core associations between MDD-PRS and depressive trajectories replicated across sites, some quantile-specific effects—most notably the initial-level effect in São Paulo and the upper-quantile trend in Porto Alegre—did not, and these site-specific patterns warrant further replication in independent admixed cohorts. Future work should extend these analyses across the full lifespan, incorporate repeated NSSI assessments to model its course, extend the cross-disorder PRS approach using multivariate frameworks (e.g., Genomic SEM) to further decompose disorder-specific from shared genetic contributions, and

continue building and validating PRS as discovery GWAS expand and diversify to improve predictive power and cross-ancestry portability.

CONCLUSIONS

This longitudinal study of a Brazilian high-risk cohort, using a discovery-replication design across two metropolitan sites, found that PRS for MDD showed relative diagnostic specificity for a lifetime MDD diagnosis, independent of psychiatric comorbidities. Longitudinally, higher MDD-PRS predicted higher initial and average levels of depressive symptoms and a faster rate of increase from childhood to young adulthood, and a greater burden of NSSI confined to individuals with lifetime MDD. Thus, the relative diagnostic specificity of PRS for MDD extends beyond categorical diagnosis to predict the developmental course of symptoms and severity, demonstrating the value of deep phenotyping in diverse, admixed populations. Future work should test mechanisms (e.g., stress reactivity, sleep, social adversity) and evaluate early-intervention utility and treatment response in diverse cohorts.

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FIGURES TITLES AND LEGENDS

Figure 1. Specificity of major depressive disorder polygenic risk score (MDD-PRS) after adjustment for general comorbidity. (A) Average marginal effect (AME; %-point change in diagnosis probability per 1-standard deviation (SD) increase in the MDD-PRS) for MDD, conduct disorder/oppositional defiant disorder (CD/ODD), attention-deficit/hyperactivity disorder (ADHD), other disorders, and anxiety, by site (Porto Alegre; São Paulo). Points = AME; horizontal bars = 95% confidence interval (CI); dashed vertical line = null. **(B)** Adjusted probability of MDD across MDD-PRS deciles by site; solid lines = predicted probability; shaded ribbons = 95% CI; annotations show p15 and p85 (15th and 85th percentiles). Models are weighted logistic and adjust for age, sex at birth, socioeconomic status (SES), and genetic principal components (PCs 1–10).

Figure 2. Longitudinal trajectory of depressive symptoms vs. major depressive disorder polygenic risk score (MDD-PRS). Columns show latent growth curve model (LGCM) parameters—Initial level (intercept), Rate of change (slope), and time-averaged modeled level (area under the curve [AUC]/time); rows show sites. Curves plot the quantile-regression (QR) coefficient $\beta(\tau)$ for standardized MDD-PRS; colored ribbons = pointwise 95% confidence interval (CI); light-gray bands = simultaneous 95% CI; horizontal solid/dashed lines = ordinary least squares (OLS) estimate and its 95% CI. Coefficients are standardized; models adjust for age, sex at birth, socioeconomic status (SES), and genetic principal components (PCs) 1–10 (weighted).

Figure 3. Association between the major depressive disorder polygenic risk score (MDD-PRS) and depressive symptoms (Short Mood and Feelings Questionnaire [SMFQ]) at Wave 2. Panels show sites. Curves are quantile-regression (QR) coefficients for standardized MDD-

PRS across SMFQ quantiles (τ); colored ribbons = pointwise 95% confidence interval (CI); light-gray bands = simultaneous 95% CI; dashed line = ordinary least squares (OLS) estimate with 95% CI. Models adjust for age, sex at birth, socioeconomic status (SES), and genetic principal components (PCs) 1–10; variables were z-standardized within each site.

Figure 4. Interaction between major depressive disorder (MDD) polygenic risk score (MDD-PRS) and lifetime MDD diagnosis in predicting nonsuicidal self-injury (NSSI). Top: main effect in the full cohort; middle: adolescents with lifetime MDD; bottom: without MDD. Dashed line = standardized ordinary least squares (OLS) estimate (95% confidence interval [CI] band); curve = quantile-regression (QR) coefficients across τ . QR is not estimable for $\tau < 0.79$ owing to zero-inflation in NSSI. Models adjust for age, sex at birth, socioeconomic status (SES), and genetic principal components (PCs) 1–10 (pooled across sites). Higher MDD-PRS predicts greater NSSI only among those with lifetime MDD (MDD $\times$ PRS interaction).

Table 1. Demographic, Socioeconomic, and Genetic Characteristics by Recruitment Site and Wave (All Results Adjusted for IPW and Oversampling, Except for N)

		Porto Alegre N W0=1104; N W1=915; N W2=891 ^{a,b}	São Paulo N W0=1059; N W1=861; N W2=799 ^{a,b}	SMD [95% CI] ^c
Age (years)	<i>Wave 0</i>	10.32 (2.03)	9.94 (1.82)	0.20 [0.12, 0.29]
	<i>Wave 1</i>	13.61 (2.05)	13.20 (1.81)	0.22 [0.12, 0.31]
	<i>Wave 2</i>	18.28 (2.13)	17.97 (1.89)	0.16 [0.06, 0.26]
Sex at birth				0.11 [0.03, 0.19]
% Female		524 (49.5%)	463 (44.0%)	
% Male		580 (50.5%)	596 (56.0%)	
SES Score	<i>Wave 0</i>	18.04 (4.60)	18.60 (4.36)	-0.13 [-0.21, -0.04]
	<i>Wave 1</i>	18.52 (4.27)	18.33 (4.32)	0.04 [-0.05, 0.14]
	<i>Wave 2</i>	23.68 (7.58)	23.95 (7.46)	-0.04 [-0.13, 0.06]
Cohort Strata				0.25 [0.16, 0.33]
Community		425 (56.6%)	411 (44.4%)	
High-risk		679 (43.4%)	648 (55.6%)	

Genetic Ancestry (%)

African	20.58 (17.05)	27.51 (14.14)	-0.45 [-0.53, -0.36]
East Asian	0.99 (1.80)	1.48 (4.63)	-0.16 [-0.24, -0.07]
European	63.13 (18.72)	60.41 (14.74)	0.16 [0.08, 0.25]
Native American	15.30 (7.63)	10.60 (4.90)	0.76 [0.67, 0.84]

SES Category

A/B (High)	<i>Wave 0</i>	178 (15.7%)	188 (18.0%)
	<i>Wave 1</i>	134 (15.4%)	141 (15.8%)
	<i>Wave 2</i>	181 (21.7%)	176 (21.1%)
C (Middle)	<i>Wave 0</i>	773 (71.2%)	778 (73.2%)
	<i>Wave 1</i>	694 (75.2%)	632 (74.8%)
	<i>Wave 2</i>	520 (63.2%)	472 (63.2%)
D/E (Low)	<i>Wave 0</i>	153 (13.1%)	93 (8.8%)

<i>Wave 1</i>	87 (9.4%)	88 (9.4%)
<i>Wave 2</i>	144 (15.1%)	106 (15.8%)

Notes:

^a Values for continuous variables are Mean (SD). Values for categorical variables are unweighted N (weighted %).

IPW = inverse-probability-of-retention weights; SES = socioeconomic status; MDD-PRS = major depressive disorder polygenic risk score; SD = standard deviation; SMD = standardized mean difference; CI = confidence interval. Wave 0 = baseline (2010–2011); Wave 1 = 4-year follow-up (2014–2015); Wave 2 = 8-year follow-up (2018–2019).

^b N values in column headers represent participants present at that wave with non-missing MDD-PRS data.

^c SMD (Standardized Mean Difference) and 95% CI (Confidence Interval) compare Porto Alegre and São Paulo for the given variable or wave. Interpretation: Negligible if $|\text{SMD}| < 0.2$; Small if $0.2 \leq |\text{SMD}| < 0.5$; Moderate if $0.5 \leq |\text{SMD}| < 0.8$; Large if $|\text{SMD}| \geq 0.8$.

Table 2. Mental Health Outcomes by Recruitment Site (All Results Adjusted for IPW and Oversampling, Except for N)

		Porto Alegre N W0=1104; N W1=915; N W2=891 ^{a,b}	São Paulo N W0=1059; N W1=861; N W2=799 ^{a,b}	SMD [95% CI] ^c
Lifetime Categorical Diagnoses^d				
Major Depressive Disorder (MDD)		211 (17.0%)	146 (11.3%)	0.15 [0.07, 0.24]
Anxiety Disorders		305 (25.2%)	236 (19.4%)	0.13 [0.05, 0.22]
Attention-Deficit/Hyperactivity Disorder (ADHD)		194 (15.7%)	124 (9.1%)	0.19 [0.10, 0.27]
Conduct Disorder (CD) or Oppositional Defiant Disorder (ODD)		154 (13.2%)	82 (6.6%)	0.21 [0.13, 0.30]
Other Psychiatric Disorders		155 (12.2%)	109 (7.5%)	0.14 [0.06, 0.23]
Dimensional Symptom Scores				
CBCL/ABCL (Harmonized) Depressive Problems (0–24)	<i>Wave</i> <i>0</i>	2.64 (3.04)	1.66 (2.40)	0.32 [0.24, 0.41]

CBCL/ABCL (Harmonized) Depressive Problems (0–24)	<i>Wave</i> <i>1</i>	2.68 (3.21)	1.86 (2.58)	0.26 [0.17, 0.35]
CBCL/ABCL (Harmonized) Depressive Problems (0–24)	<i>Wave</i> <i>2</i>	3.47 (3.73)	3.01 (3.58)	0.12 [0.02, 0.22]
SMFQ Depressive Symptoms (0–26)	<i>Wave</i> <i>2</i>	5.21 (5.94)	5.09 (6.05)	0.02 [-0.09, 0.12]
DSHI-9 Nonsuicidal Self-Injury (0–54)	<i>Wave</i> <i>2</i>	0.71 (2.84)	0.74 (2.96)	-0.01 [-0.11, 0.09]

Notes:

See Supplementary Table S2 for a complete table with all individual diagnoses.

IPW = inverse-probability-of-retention weights; MDD = major depressive disorder; ADHD = attention-deficit/hyperactivity disorder; CD = conduct disorder; ODD = oppositional defiant disorder; CBCL = Child Behavior Checklist; ABCL = Adult Behavior Checklist; SMFQ = Short Mood and Feelings Questionnaire; DSHI-9 = Deliberate Self-Harm Inventory, 9-item version; SD = standard deviation; SMD = standardized mean difference; CI = confidence interval. Wave 0 = baseline (2010–2011); Wave 1 = 4-year follow-up (2014–2015); Wave 2 = 8-year follow-up (2018–2019).

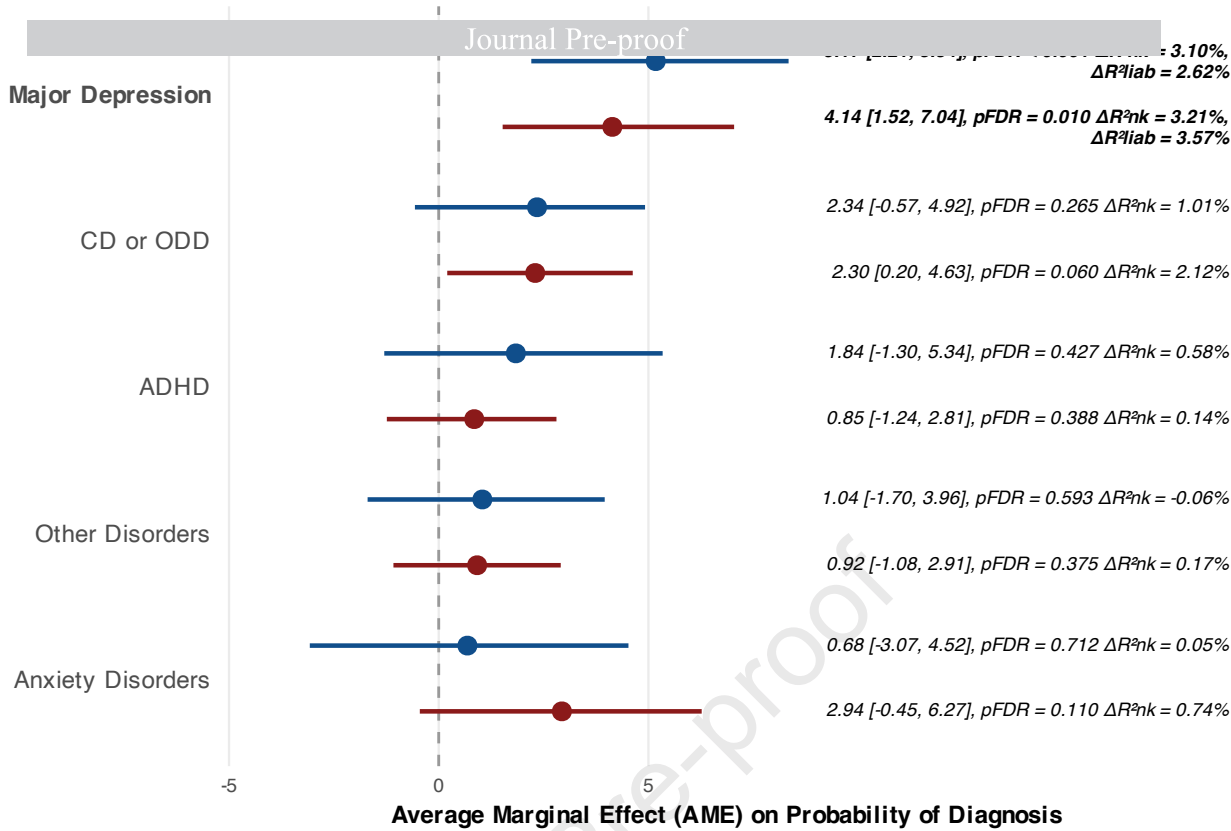
^a Values are Mean (SD) for dimensional scores. For categorical diagnoses, values are unweighted N (weighted %).

^b N values in column headers represent participants in the analytic sample for that wave.

^c SMD (Standardized Mean Difference) and 95% CI (Confidence Interval) compare Porto Alegre and São Paulo. Interpretation: Negligible if $|SMD| < 0.2$; Small if $0.2 \leq |SMD| < 0.5$; Moderate if $0.5 \leq |SMD| < 0.8$; Large if $|SMD| \geq 0.8$.

^d Lifetime categorical diagnoses are based on data from any wave, reported for the Wave 0 analytic sample.

A



B

